TECHNOVA RETAIL

Sales & Employee Performance Analysis

Data Analytics Capstone Project

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Tools: Microsoft Excel | MySQL | Power BI | DAX | Power Query

Project Type: Sales & Employee Performance Analytics

Report Overview

This report documents the end-to-end analysis of TechNova Retail's sales and employee performance data, including data preparation, Excel analysis, SQL querying, Power BI modeling, visualization, and business insights.

1. Executive Summary

TechNova Retail required a data-driven view of sales and employee performance across its regions. Sales data was provided in Microsoft Excel and employee performance data in MySQL. Excel supported spreadsheet analysis, MySQL supported structured queries, and Power BI supported data modeling, DAX, visualization, interactive filtering, and drill-through. The completed sales analysis recorded **■5,400** in total sales, with **Washing Machine** identified as the highest-selling product.

2. Business Problem

Sales and employee-performance information were stored separately. Management required a consolidated analytical solution to evaluate regional and product performance, estimated profit, employee target attainment, bonus eligibility, and sales trends.

3. Project Objectives

The project aimed to calculate regional sales, identify the highest-selling product, estimate profit using the specified 20% margin, perform employee lookups and target analysis, query employee performance in MySQL, and build an interactive Power BI dashboard with slicers and regional drill-through.

4. Data Sources & Dataset Description

Sales Data — Excel: OrderID, CustomerName, Product, Category, Quantity, UnitPrice, Profit, OrderDate, Region, SalesAmount, and ActualSales.
Employee Performance — MySQL: EmployeeID, EmployeeName, Region, SalesTarget, ActualSales, and BonusEligibility.

5. Analytical Approach

The workflow followed data collection → data preparation → analysis → data modeling → DAX → visualization → business interpretation. Excel was used for spreadsheet analysis, MySQL for structured queries, and Power BI for integrated reporting.

6. Data Preparation & Cleaning

The datasets were checked for missing values, data types, duplicate records, consistency, and date readiness. Employee performance records were reviewed to prevent duplicate rows from inflating performance totals. The final employee dataset contains 5 unique employees.

7. Excel Analysis

7.1 Total Sales per Region

A Pivot Table summarized SalesAmount by Region. Total sales across the 5 sales transactions were **■5,400**.

Region	Total Sales
North	■1,900
West	■1,600
East	■1,200
South	■700

Finding: North generated the highest sales in the Excel sales dataset at **■1,900**.

7.2 Highest-Selling Product

Finding: Washing Machine was the highest-selling product, generating **■1,600**.

7.3 Profit Calculation

Using the required 20% profit margin, estimated total profit is **■1,080**. Profit was calculated as SalesAmount × 20%.

7.4 Employee Performance Lookup

XLOOKUP/VLOOKUP logic was used to match employee identifiers with corresponding performance information and support target-versus-actual analysis.

7.5 Conditional Formatting

Employees with ActualSales below SalesTarget were highlighted using conditional formatting.

7.6 Excel Dashboard

The Excel dashboard summarizes total sales, product performance, regional performance, profit, and employee performance.

8. MySQL Analysis

8.1 Employees Exceeding Target by More Than 10%

```
SELECT EmployeeID, EmployeeName, Region, SalesTarget, ActualSales
FROM employees_performance
WHERE ActualSales > SalesTarget * 1.10;
```

The query identified **1** employee(s) whose ActualSales exceeded SalesTarget by more than 10%.

8.2 Bonus-Eligible Employees

```
SELECT EmployeeID, EmployeeName, Region, ActualSales, BonusEligibility
FROM employees_performance
WHERE BonusEligibility = 'Yes';
```

The query identified **3** bonus-eligible employee(s).

8.3 Region With the Highest Sales Performance

```
SELECT Region, SUM(ActualSales) AS TotalSales
FROM employees_performance
GROUP BY Region
ORDER BY TotalSales DESC;
```

After cleaning the employee data, **North** recorded the highest employee actual-sales total at **■5,200**.

9. Power BI Data Modeling

The Excel sales data and MySQL employee performance data were brought into Power BI. The model was reviewed for appropriate relationships, correct data types, valid filter behavior, and duplicate-related distortion.

10. Power BI Dashboard

The Power BI dashboard provides management with headline metrics and interactive analysis. It includes KPI cards, sales by region, top products, employee performance, monthly sales trend, slicers, and a Region Details drill-through page. Headline results include total sales of **■5,400**, total estimated profit of **■1,080**, total quantity of **7**, and average sales of **■1,080**.

10.1 Employee Target Achievement

3 of 5 employees (60.0%) met or exceeded their sales targets. Employees below target were highlighted for management attention.

10.2 Drill-Through Analysis

The Region Details page allows users to select a region and inspect employee-level target, actual-sales, and performance information.

10.3 Data Refresh

The solution uses Excel and MySQL as source systems. In a production environment, scheduled refresh would depend on Power BI Service, credentials, gateway configuration, and source accessibility.

11. Key Business Insights

Overall Sales: The sales dataset recorded **■5,400** across 5 transactions.

Regional Performance: North generated the highest sales in the Excel sales data at **■1,900**. In the cleaned employee performance data, North also had the highest actual-sales total at **■5,200**.

Product Performance: Washing Machine was the highest-selling product with **■1,600** in sales.

Estimated Profit: The assumed 20% margin produced **■1,080** in estimated profit.

Target Achievement: 3 of 5 employees met or exceeded target (60.0%).

Exceptional Performance: 1 employee(s) exceeded target by more than 10%.

Bonus Eligibility: 3 employee(s) were marked as bonus eligible.

Dataset Limitation: The dataset contains only 5 sales transactions and 5 employee records. The results demonstrate the analytical workflow but should not be interpreted as a statistically representative view of a large retail operation.

12. Challenges & Solutions

Database connection: Additional connector components were required before the employee data could be accessed.
Data quality: Duplicate employee records could inflate performance totals, so the final analysis used 5 unique employee records.
Histogram: Quantity bins were used to represent the required distribution.
Dashboard design: The initial design was improved through stronger spacing, typography, KPI presentation, consistency, and reduced visual clutter.

13. Project Deliverables

Excel: cleaned analysis, Pivot Tables, profit calculation, lookup analysis, conditional formatting, and dashboard. **MySQL:** employee table, SQL queries, results, and screenshots. **Power BI:** data model, DAX measures, dashboard, KPI cards, charts, slicers, conditional formatting, and drill-through. **Documentation:** report, README, screenshots, datasets, and final dashboards.

14. GitHub Repository Structure

TechNova-Sales-Employee-Analysis/ ■■■ README.md ■■■ Dataset/ ■■■ Excel/ ■■■ SQL/ ■■■ PowerBI/ ■■■ Report/ ■■■ TechNova_Analysis_Report.pdf

15. Conclusion

The TechNova project demonstrates an end-to-end analytics workflow using Excel, MySQL, Power BI, DAX, and Power Query. The analysis found **■5,400** in total sales, identified **Washing Machine** as the highest-selling product, and identified **North** as the leading region in both the Excel sales data and cleaned employee actual-sales analysis. The project demonstrates practical skills in data cleaning, SQL, data modeling, DAX, visualization, dashboard development, and business insight communication.